

Draft - expert review pending

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Topic 2.3 - Solid and hazardous waste

A real classroom situation

A trainee teacher on practicum in Karachi walks past the school gate at 7 a.m. The sweeper is burning the day's rubbish in a corner of the lane: paper, plastic bags, food scraps and, on top, a cardboard box with a red stripe that has come loose from the clinic next door. The smoke drifts across the assembly line. The trainee has just read about the waste hierarchy, and she now sees three failures in one heap: waste that should have been reduced, waste that should have been separated, and clinical waste that should never have been in the lane at all.

This topic gives you the principles that heap breaks: how a city manages its solid waste, and how the hazardous part, hospital and industrial waste above all, is kept apart.

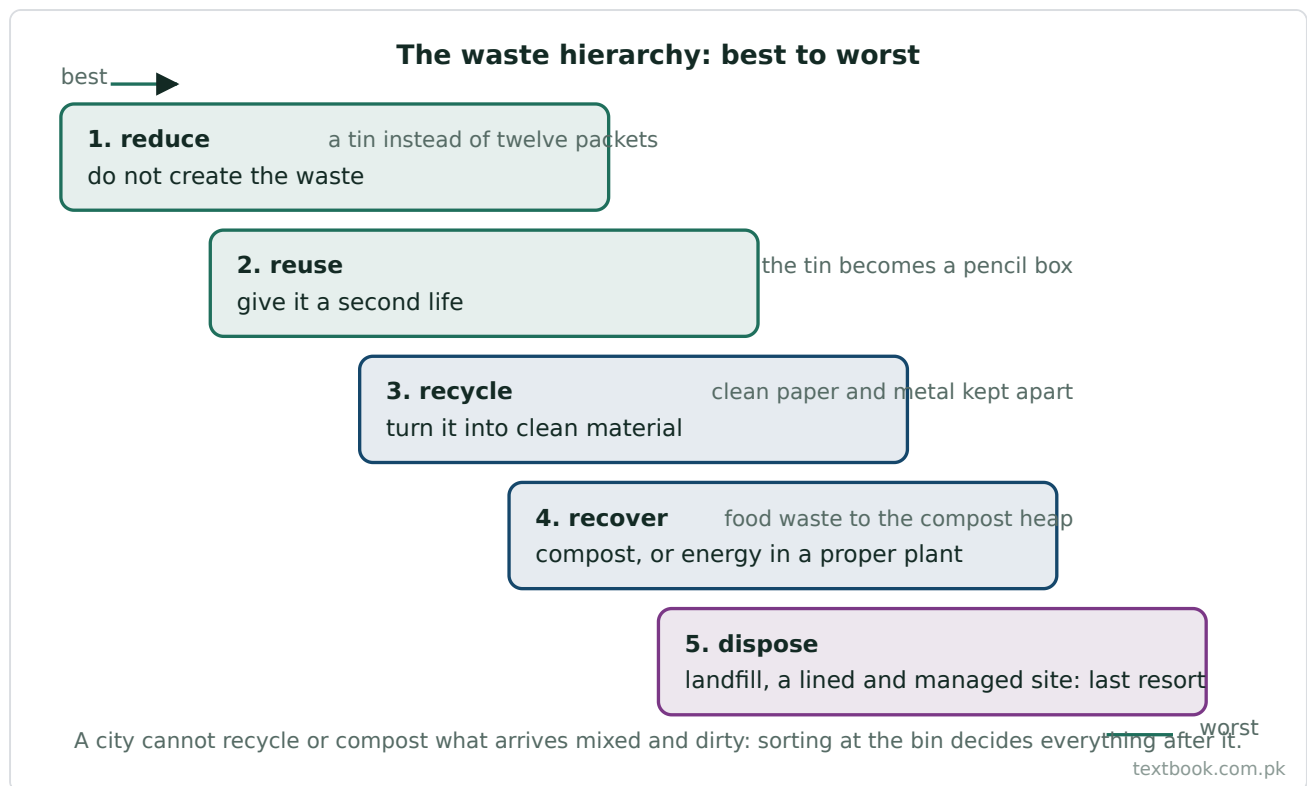
Explanation

The principles of municipal solid waste management

Municipal solid waste is the everyday waste of households, shops, schools and offices: food scraps, paper, plastic, glass, metal, garden waste and dust.

Managing it is a system with five stages: generation, storage at source, collection, transport, and disposal or recovery. A city fails at whichever stage it neglects; the burning heap at the school gate is a storage and collection failure before it is anything else.

The ordering principle for the whole system is the waste hierarchy, a staircase from best to worst.



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- Reduce comes first because waste never made needs no bin, no truck and no landfill. This is Unit 1's strongest step, restated for a city.
- **Reuse** gives an object a second life without reprocessing it.
- Recycle turns material into material, and depends on the recovery chain buying clean, separated stock: paper, cardboard, metal, some plastics.
- **Recover** extracts what is left, compost for organic waste, or energy where a proper facility exists.
- **Dispose** is the last resort, and even then a lined, managed site, not an open heap.

The stakes are real. A study of Pakistan's metropolitan dumpsites estimated decades of methane emissions and leachate from Lahore's disposal systems alone at about 108 million tonnes of carbon-dioxide equivalent, and leachate, the liquid that drains from waste, is a direct threat to groundwater (Ilmas et al., 2026). The WHO reports that 44 per cent of household wastewater worldwide is discharged without safe treatment (WHO, 2024); the solid waste story is the

same shape. Segregation at source, keeping the streams apart while they are still clean, is the step that makes every later step possible.

Hazardous waste management: hospital and industrial waste

Hazardous waste is waste whose properties make it dangerous to people or the environment: infectious, toxic, flammable, corrosive or sharp. The cardinal rule is that it never mixes with the ordinary stream, because a bag of hospital waste in a municipal heap turns the whole heap hazardous.

Hospital, or clinical, waste sorts into two streams. The **general** stream, offices and kitchens, behaves like municipal waste. The **risk** stream is what the rules exist for: sharps (needles, blades), infectious waste (dressings, laboratory cultures), pathological waste, pharmaceutical waste and chemical waste. Its handling rules are simple and strict: segregate at the point of generation into colour-coded containers, a sharps container that cannot be reopened, and a covered infectious bin; label and date them; store briefly in a secured area; and hand them to a licensed handler for treatment, autoclaving or incineration, before final disposal. The people most at risk when the rules fail are the sweeper, the waste picker and the nurse, in that order.

Industrial hazardous waste, solvent sludges, electroplating liquids, chemical drums, follows the same logic with its own paperwork: characterise the waste, minimise it, store it in contained, labelled space, and transfer it with a manifest so its path can be traced from generator to licensed disposer.

Two hazardous streams, two rule sets

Stream	Typical items	Why hazardous	Handling rule
Hospital clinical waste	sharps (needles, blades); infectious dressings; laboratory cultures; expired medicines	infection and injury: a needle can pass disease straight through the skin	segregate at the point of use into colour-coded bins; sealed sharps box; licensed treatment
Industrial hazardous waste	solvent sludges; electroplating liquids; chemical drums; paint and acid residues	toxic, flammable or corrosive; can poison groundwater for decades	characterise; minimise; store contained and labelled; transfer with a manifest, licensed
The cardinal rule	hazardous waste never mixes with the ordinary stream: one clinical bag makes a whole municipal heap hazardous. Most at risk when rules fail: the sweeper, the picker, the nurse.		

Segregate at generation, label and date, store briefly and securely, hand to licensed treatment.

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A worked example. Return to the burning heap. The paper and food scraps belong lower on the hierarchy, separated for the recovery chain and for compost. The plastic bags are a reduce failure from the day they were issued. And the red-striped box is the sharpest lesson: clinical waste in the municipal stream, burning at a school gate, exposing the sweeper, the pupils and the whole lane to smoke from infectious material. Three streams, three rules, one heap that broke all of them.

Common misconception. "Waste management means collection." Collection is one stage of five, and it is not the first. The first is what household and shop decide to generate and separate. A city cannot recycle, compost or safely dispose what arrives mixed and dirty; the sorting that happens at the bin decides the fate of everything after it.

Activity: Bin audit, three streams

Work in groups of three. With permission, examine one day of your classroom block's waste (avoid touching anything: use tongs or gloved hands and a spread sheet). Sort it into three streams on the spread sheet: recoverable (clean paper, cardboard, metal), organic (food, garden), and residual (the rest, including

anything soiled). Weigh or count each stream, then answer: what share never needed to be generated? Allow 30 minutes, including washing hands afterwards.

Check your understanding

1. Name the five steps of the waste hierarchy in order, from best to worst. *(Remember)*
2. Why does reduce sit above recycle, and what does recycling depend on that reduce does not? *(Understand)*
3. A clinic's dressings end up in the municipal bin. Name the rule that was broken and two groups of people put at risk. *(Apply)*
4. Your city has money for one change: a fleet of trucks, or a segregation campaign with bins. Use the hierarchy and the dumpsite evidence to argue for one. *(Analyze)*

Summary

Municipal solid waste runs through five stages, generation, storage, collection, transport, disposal or recovery, ordered by the waste hierarchy: reduce, reuse, recycle, recover, dispose. Pakistan's open dumpsites show the cost of skipping the hierarchy in methane and leachate. Hazardous waste, hospital risk waste and industrial waste above all, must never enter the municipal stream: segregate at the point of generation, label, store briefly and securely, and transfer to licensed treatment. The people the rules protect first are the workers who handle the waste with their hands.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can name the five stages of the waste system and the five steps of the hierarchy.
- ☐ I can explain why segregation at source decides everything downstream.
- ☐ I can list the hospital risk-waste categories and the handling rule for each.

☐ I can read a waste heap and name the failures it shows, as the worked example did.

Try this at your practicum school

With your cooperating teacher's agreement, run the two-bin classroom separation from Unit 1 for one week, then extend it by one stream: a covered bin for anything wet or soiled, kept out of the recoverable bin. Count the three streams at week's end and report the shares. Note one thing the school could change so that the residual stream shrinks.

Summative task

Write 170 to 200 words on the following.

Your school produces a mixed waste heap at its gate. Write a one-page plan outline, in your own words, for moving the school up the waste hierarchy: name the first change, the second change, who must agree to each, and how you would show the head teacher it is working by the end of the term.

Mini-rubric

Criterion	What a strong response does
Hierarchy applied (Apply)	Orders the changes from reduce downward, not a random list
Feasibility (Understand)	Names who agrees and what each change costs in effort, realistically
Evidence plan (Analyze)	Defines a measurable end-of-term check, counts or shares, not a feeling
Clarity and register	Plain sentences for a head teacher; within the word count

Further reading

- Ilmas, B., Khalid, S., Ijaz, M., & Hussain, I. (2026). Emissions and leachate profile of MSW disposal sites of metropolitan cities of Pakistan using LandGEM model. *Carbon Balance and Management*, 21(1).
<https://doi.org/10.1186/s13021-026-00403-x>
- World Health Organization. (2024). *Sanitation* (fact sheet):
<https://www.who.int/news-room/fact-sheets/detail/sanitation>
- Holland, W. W. (Ed.). *Oxford Textbook of Public Health*. (guide-recommended book's closest real work, cited by reference only; flagged per D-2026-0001)

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